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DSA0602-Data Handling And Visualization

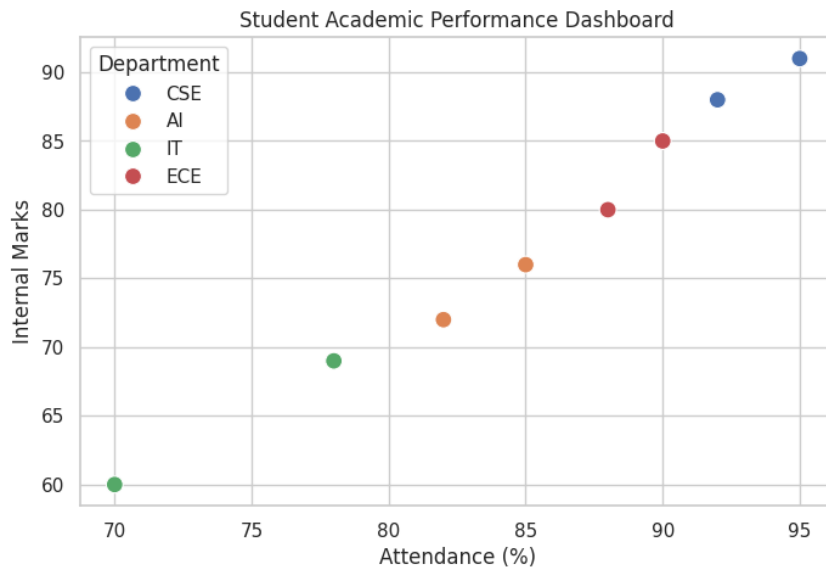
Class Activity 1

Scenario 1: Student Academic Performance Dashboard

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")
df = pd.DataFrame({
    "Student_ID": ["S1","S2","S3","S4","S5","S6","S7","S8"],
    "Attendance": [92, 85, 78, 88, 95, 70, 82, 90],
    "Internal_Marks": [88, 76, 69, 80, 91, 60, 72, 85],
    "Department": ["CSE", "AI", "IT", "ECE", "CSE", "IT", "AI", "ECE"]
})
plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x="Attendance", y="Internal_Marks", hue="Department",
s=100)
plt.title("Student Academic Performance Dashboard")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.legend(title="Department")
plt.show()
```

OUTPUT:



Scenario 2: Electric Vehicle Market Analysis

CODE:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
sns.set(style="whitegrid")
```

```
df = pd.DataFrame({  
    "Vehicle_Model": ["EV1", "EV2", "EV3", "EV4", "EV5", "EV6"],  
    "Battery_Capacity": [30, 40, 50, 60, 75, 90],  
    "Price": [8, 10, 14, 18, 24, 30], # in lakhs  
    "Driving_Range": [200, 250, 320, 380, 450, 520],  
    "Manufacturer": ["Tata", "Mahindra", "Tata", "MG", "Hyundai", "BYD"]  
})
```

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(  
    data=df,  
    x="Battery_Capacity",
```

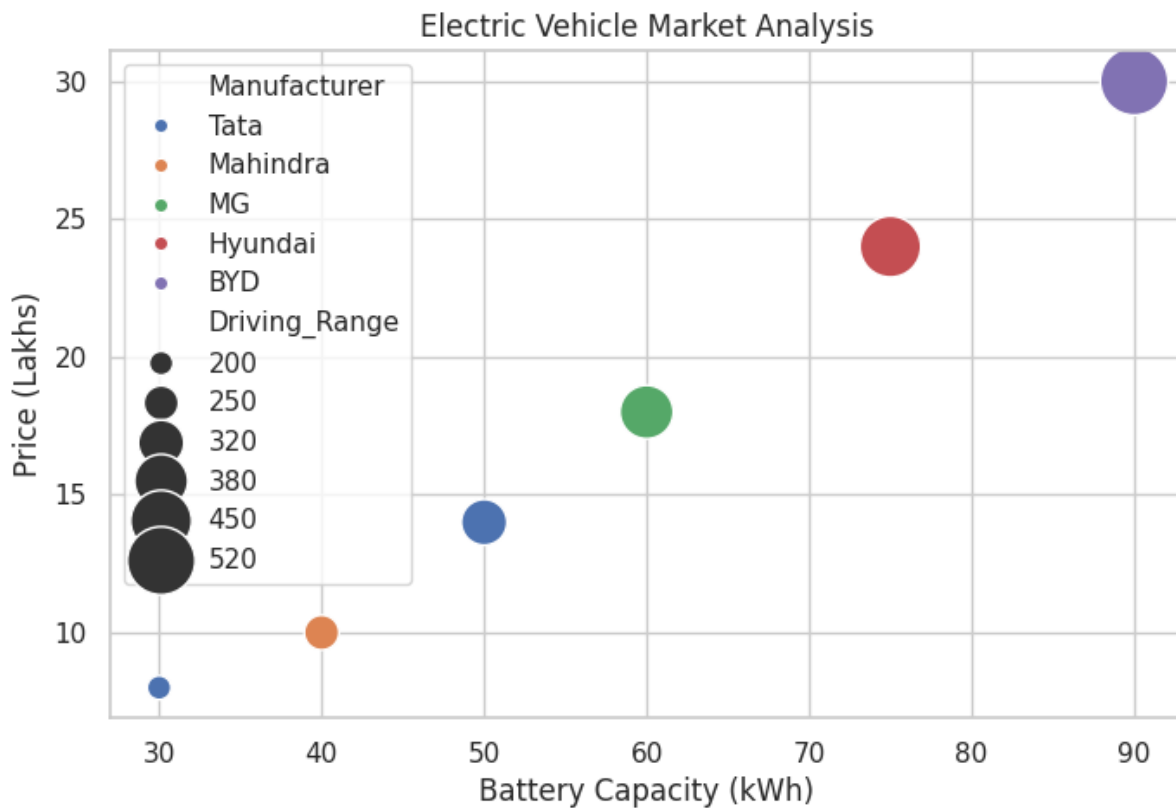
```

y="Price",
size="Driving_Range",
hue="Manufacturer",
sizes=(100, 800)
)

plt.title("Electric Vehicle Market Analysis")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price (Lakhs)")
plt.show()

```

OUTPUT:



Scenario 3: Hospital Patient Admission Analysis

CODE:

```

import pandas as pd
import matplotlib.pyplot as plt

```

```
df = pd.DataFrame({  
    "Department": ["Cardiology", "Orthopedics", "Neurology", "Pediatrics",  
    "General"],  
    "Number_of_Patients": [120, 90, 70, 110, 150],  
    "Average_Treatment_Cost": [5000, 4000, 6500, 3000, 2500]  
})
```

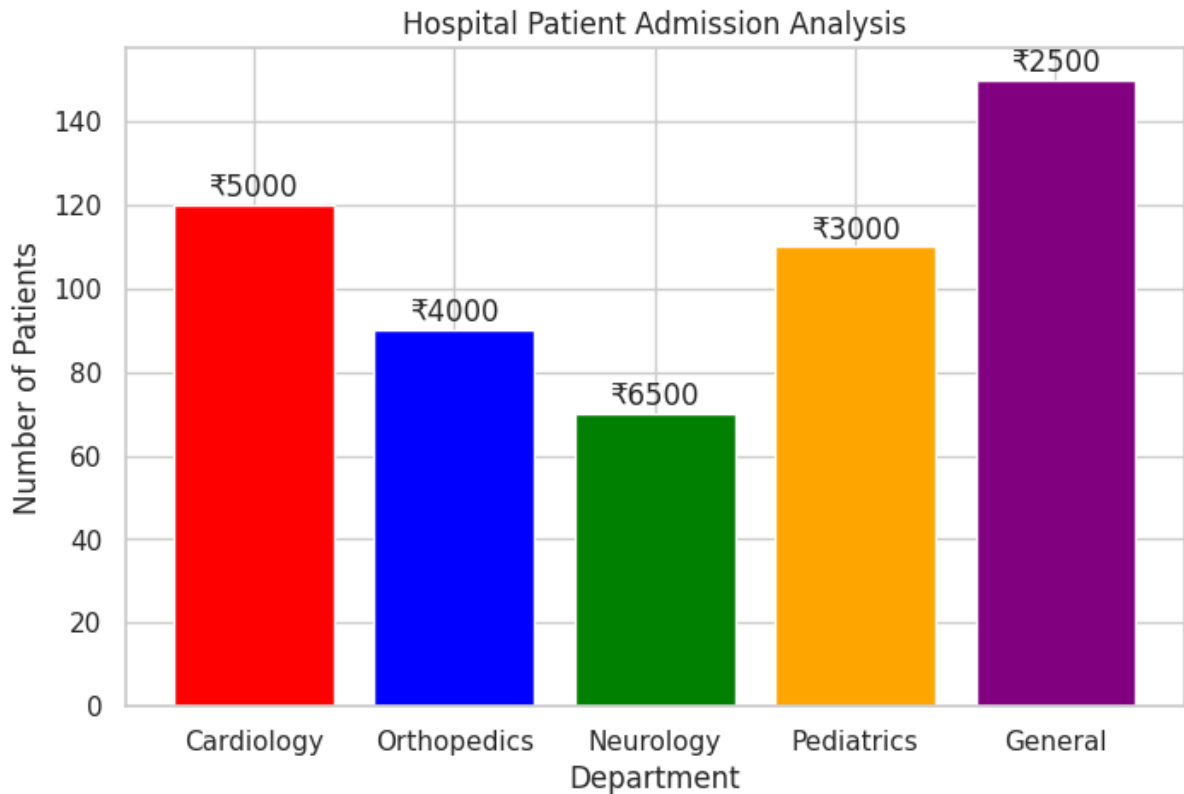
```
plt.figure(figsize=(8,5))  
bars = plt.bar(df["Department"], df["Number_of_Patients"],  
    color=["red","blue","green","orange","purple"])
```

```
plt.title("Hospital Patient Admission Analysis")  
plt.xlabel("Department")  
plt.ylabel("Number of Patients")
```

```
for bar, cost in zip(bars, df["Average_Treatment_Cost"]):  
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height()+2,  
        f"₹{cost}", ha='center')
```

```
plt.show()
```

OUTPUT:



Scenario 4: Weather Monitoring System

CODE:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df = pd.DataFrame({
    "Date": ["2026-07-01", "2026-07-02", "2026-07-03", "2026-07-04", "2026-07-05"]*3,
    "Temperature": [32,33,31,34,35, 28,29,30,31,30, 35,36,34,37,38],
    "City": ["Chennai"]*5 + ["Bangalore"]*5 + ["Delhi"]*5
})
```

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
plt.figure(figsize=(10,5))
```

```
cities = df["City"].unique()
```

```
line_styles = ["-", "--", ":"]
```

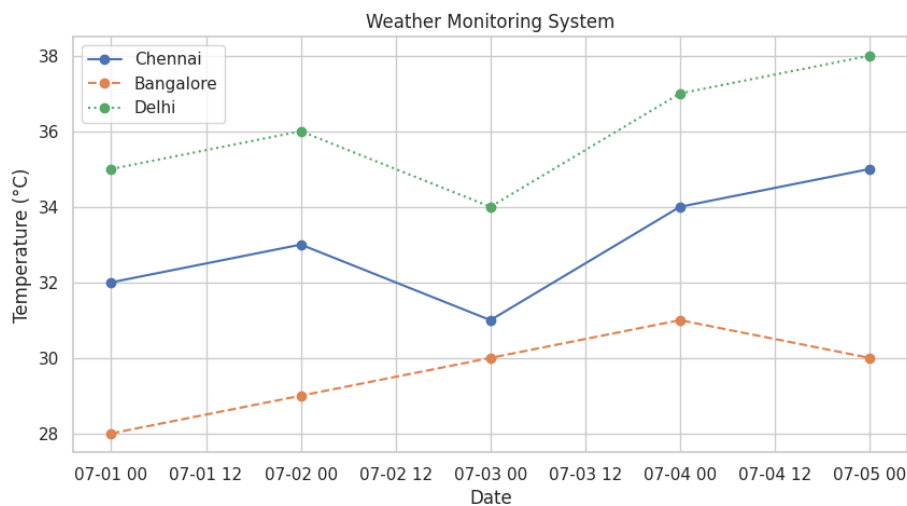
```

for city, style in zip(cities, line_styles):
    city_data = df[df["City"] == city]
    plt.plot(city_data["Date"], city_data["Temperature"],
             label=city, linestyle=style, marker='o')

plt.title("Weather Monitoring System")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
plt.legend()
plt.show()

```

OUTPUT:



Scenario 5: Supermarket Sales Analysis

CODE:

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")

df = pd.DataFrame({
    "Month":
["Jan", "Jan", "Jan", "Jan", "Feb", "Feb", "Feb", "Feb", "Mar", "Mar", "Mar", "Mar"],
    "Product_Category": ["Snacks", "Beverages", "Snacks", "Beverages"]*3,

```

```

    "Sales_Amount":
[20000,15000,18000,16000,22000,17000,21000,18000,25000,19000,23000,20000],

    "Branch": ["A","A","B","B"]*3
})

plt.figure(figsize=(10,5))

sns.barplot(data=df, x="Month", y="Sales_Amount", hue="Branch")

plt.title("Supermarket Sales Analysis")

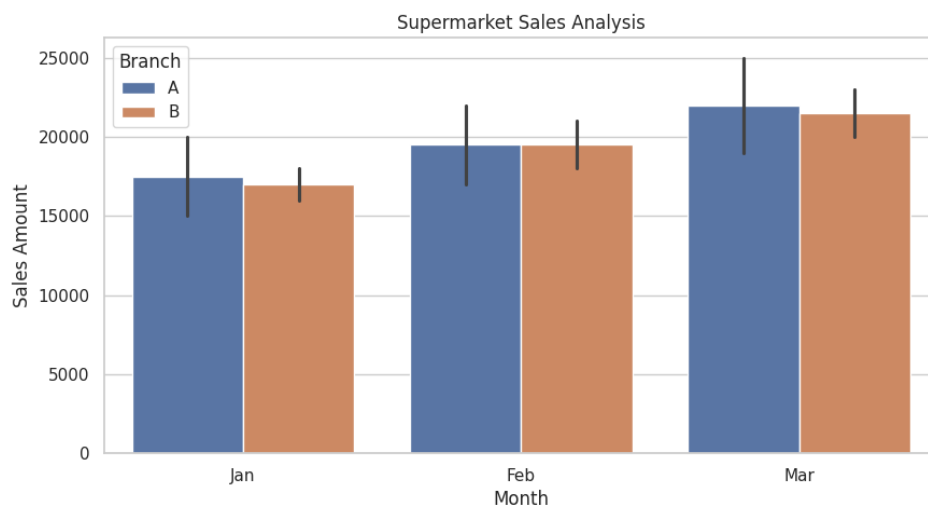
plt.xlabel("Month")

plt.ylabel("Sales Amount")

plt.show()

```

OUTPUT:



Scenario 6: Employee Salary Distribution

CODE:

```

import pandas as pd

import matplotlib.pyplot as plt

df = pd.DataFrame({

    "Employee_ID": ["E1","E2","E3","E4","E5","E6","E7","E8","E9","E10"],

    "Salary": [25000, 30000, 28000, 35000, 40000, 45000, 50000, 52000, 60000, 70000],

    "Experience": [1,2,2,3,4,5,6,7,8,10],

    "Department":
["HR","IT","IT","Sales","Finance","IT","Sales","HR","Finance","Management"]

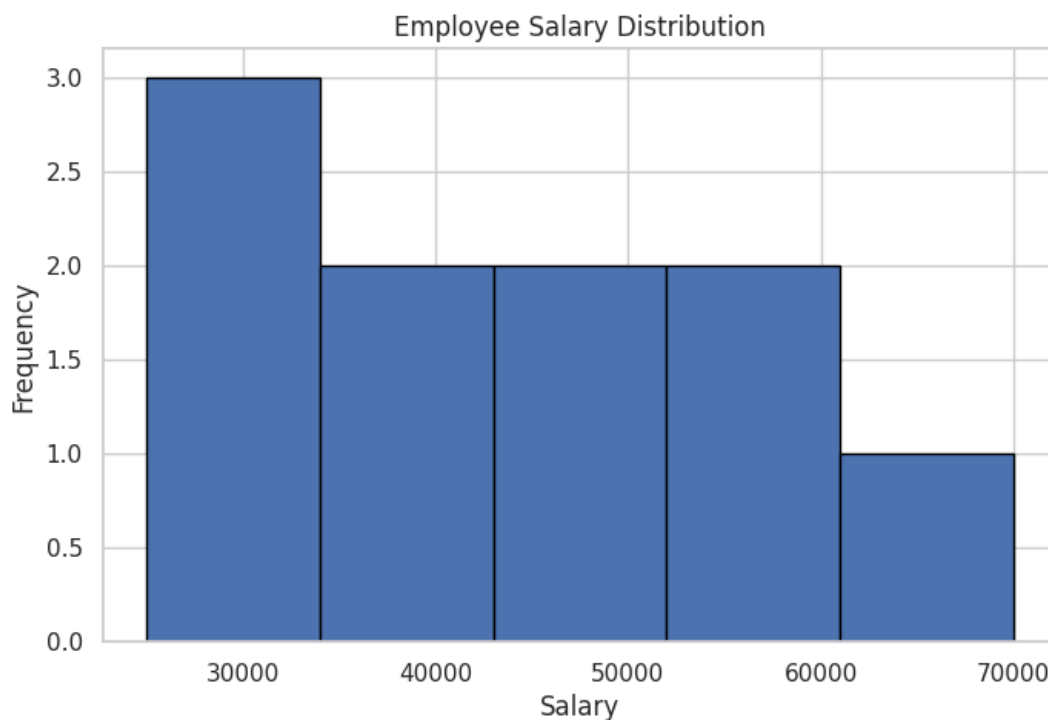
```

```

})
plt.figure(figsize=(8,5))
plt.hist(df["Salary"], bins=5, edgecolor="black")
plt.title("Employee Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Frequency")
plt.show()

```

OUTPUT:



Scenario 7: Smartphone Market Comparison

CODE:

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

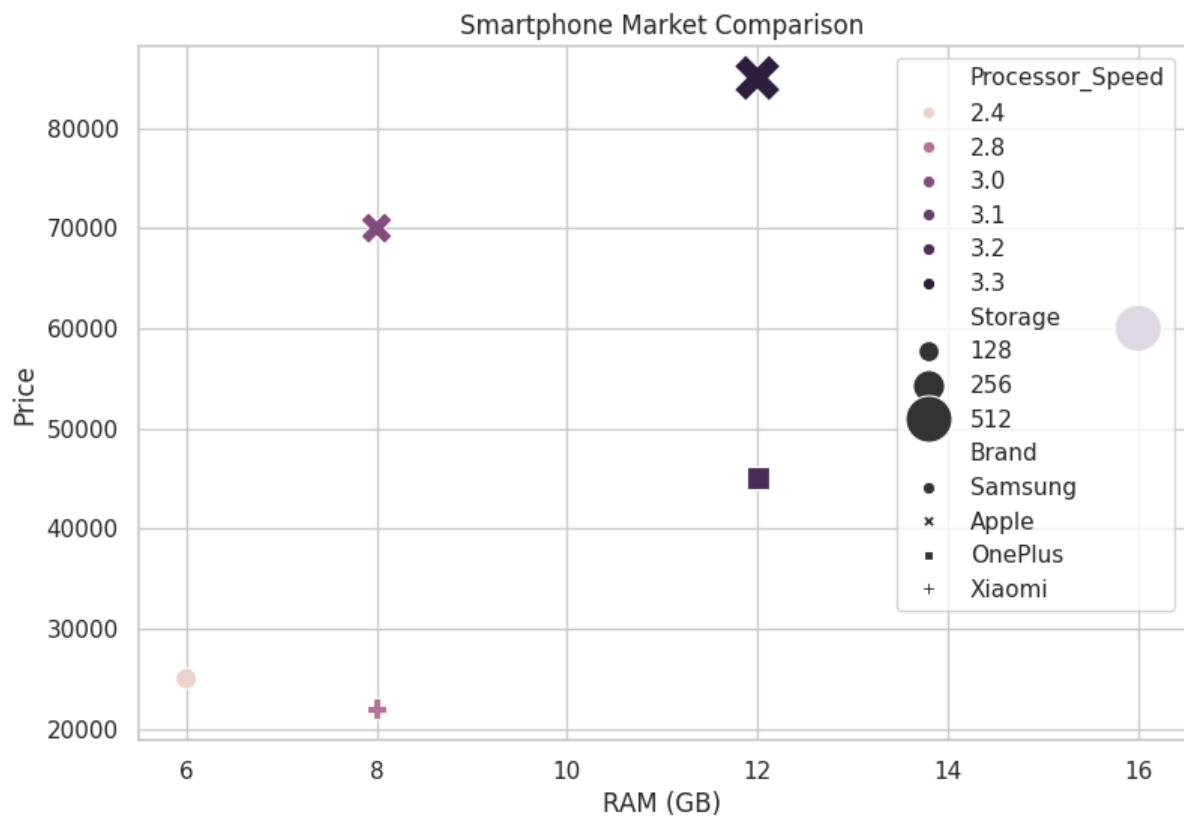
sns.set(style="whitegrid")

df = pd.DataFrame({
    "Brand": ["Samsung", "Apple", "OnePlus", "Xiaomi", "Samsung", "Apple"],

```

```
"RAM": [6, 8, 12, 8, 16, 12],  
"Processor_Speed": [2.4, 3.0, 3.2, 2.8, 3.1, 3.3],  
"Price": [25000, 70000, 45000, 22000, 60000, 85000],  
"Storage": [128, 256, 256, 128, 512, 512]  
})
```

```
plt.figure(figsize=(9,6))  
sns.scatterplot(  
    data=df,  
    x="RAM",  
    y="Price",  
    hue="Processor_Speed",  
    size="Storage",  
    style="Brand",  
    sizes=(100, 500)  
)  
plt.title("Smartphone Market Comparison")  
plt.xlabel("RAM (GB)")  
plt.ylabel("Price")  
plt.show()  
OUTPUT:
```



Scenario 8: Online Movie Rating Analysis

CODE:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
sns.set(style="whitegrid")
```

```
df = pd.DataFrame({
    "Genre": ["Action", "Comedy", "Drama", "Horror", "Sci-Fi"]*3,
    "Average_Rating": [7.8, 6.9, 8.2, 6.5, 8.0,
                       8.1, 7.2, 8.4, 6.8, 8.3,
                       7.9, 7.0, 8.5, 6.6, 8.1],
    "Number_of_Reviews": [500,400,650,300,450,
                          520,420,700,320,480,
                          530,410,720,310,500],
```

```

    "Release_Year": [2024]*5 + [2025]*5 + [2026]*5
})

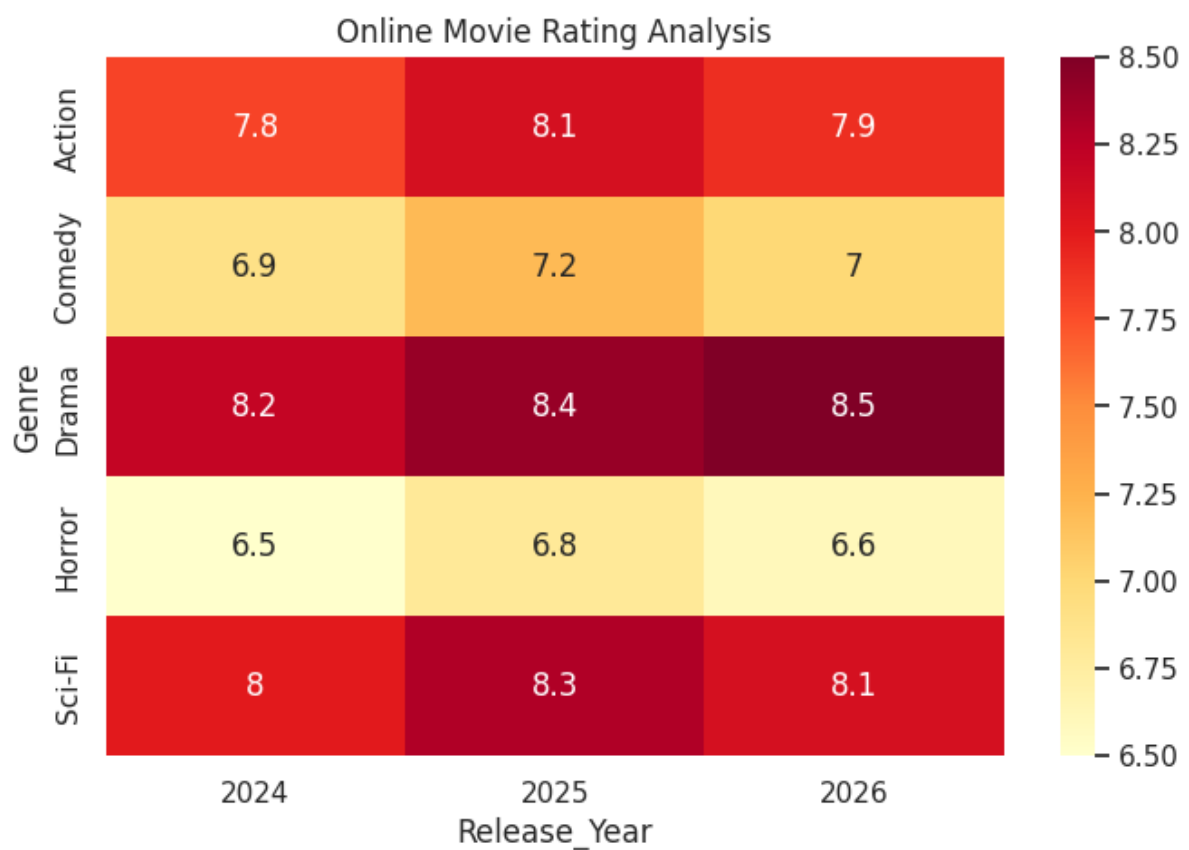
pivot_df = df.pivot(index="Genre", columns="Release_Year",
values="Average_Rating")

plt.figure(figsize=(8,5))
sns.heatmap(pivot_df, annot=True, cmap="YlOrRd")

plt.title("Online Movie Rating Analysis")
plt.show()

```

OUTPUT:

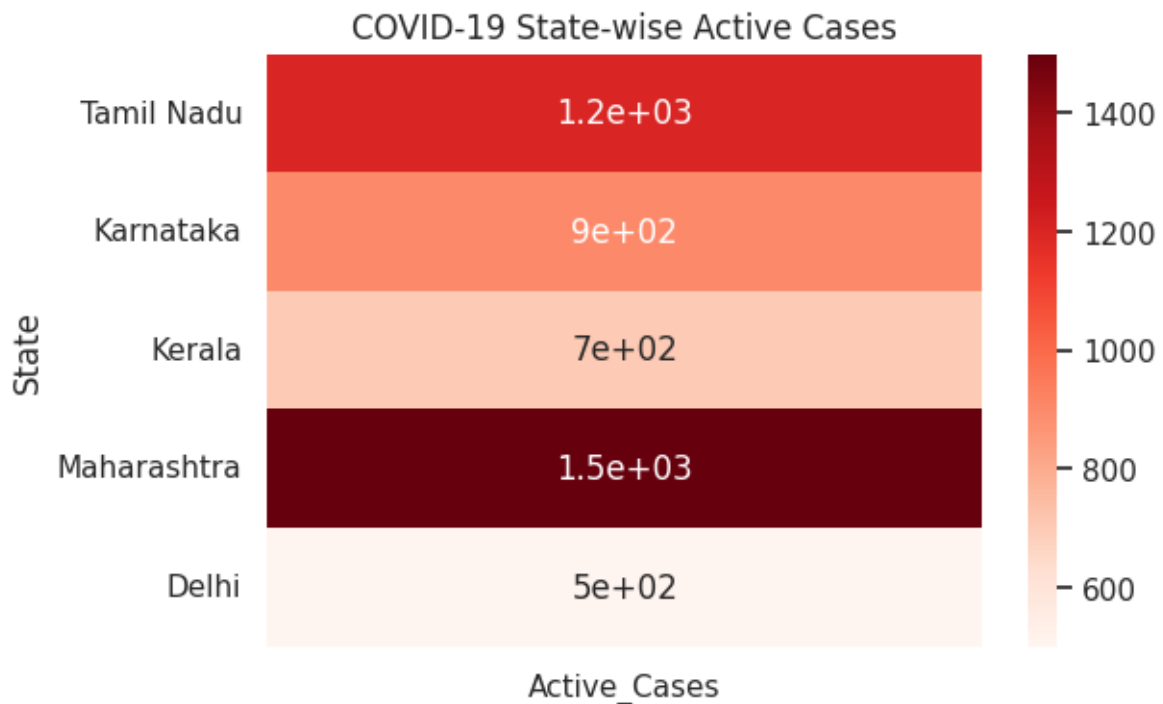


Scenario 9: COVID-19 State-wise Analysis

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")
df = pd.DataFrame({
    "State": ["Tamil Nadu", "Karnataka", "Kerala", "Maharashtra", "Delhi"],
    "Active_Cases": [1200, 900, 700, 1500, 500],
    "Vaccination_Percentage": [88, 85, 90, 84, 92],
    "Population": [77000000, 68000000, 35000000, 124000000, 19000000]
})
tile_data = df.set_index("State")[["Active_Cases"]]
plt.figure(figsize=(6,4))
sns.heatmap(tile_data, annot=True, cmap="Reds")
plt.title("COVID-19 State-wise Active Cases")
plt.show()
```

OUTPUT:



Scenario 10: Iris Flower Classification

CODE:

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")
df = sns.load_dataset("iris")
plt.figure(figsize=(9,6))
sns.scatterplot(
    data=df,
    x="sepal_length",
    y="petal_length",
    hue="species",
    style="species",
    size="sepal_width",
    sizes=(40, 200)
)
plt.title("Iris Flower Classification")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.show()
```

OUTPUT:

